Code availability is not enough: A schema-based representation for comprehensible and extensible data analysis

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ABSTRACT
Current reproducibility practices have greatly improved our ability to replicate published results, but they do not necessarily make data analyses easier to understand. Audience seeking to comprehend an analysis need to piece together information scattered across manuscripts and code scripts to understand the sequence of analytical steps and the rationale behind them, with some details may not explicitly record in either source. Exploring alternative analytical choices is even more challenging, as it requires understanding and modifying code that may be unfamiliar and consider how it would impact upstream and downstream code. To address these challenges, we propose a schema that represents data analyses as a collection of analytical decisions, together with their justification, inputs, and outputs. Coupled with large language models (LLMs), the schema can be inferred from natural language descriptions and translated into analysis scripts. The structured representation also enables exploration of alternative analytical choices, allowing multiple analytical workflows to be generated and compared. We demonstrate the proposed workflow using the many-analysts football study, where LLMs create the schema from the decision choices in team reports, generate corresponding analysis scripts, and explore alternative random effect specifications. As an intermediate representation between the manuscript and code scripts, our schema makes analyses more comprehensible and extendable from a reader's perspective.

KEYWORDS
Reproducibility; Large Language Models (LLMs); Analytical decisions; Many-analysts study

1. Introduction

When we want to understand a published data analysis, beyond simply being able to re-run it, there is currently nothing we can use directly for this purpose. Code and data availability, now standard practice following a broader push for reproducible research (Peng, 2011; Wrobel et al., 2024), guarantee that a result can be regenerated, but they

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say nothing about whether a reader can track the decisions the authors made, understand the rationale behind them, or comfortably modify the code to explore alternative choices.

Part of the difficulty is that code and the applied researchers who write it operate at different levels: code specifies precisely what a computer should do, while a researcher reasons about what decision that computation represents in the context of their analysis, a gap also identified in prior work on authoring statistical models from conceptual relationships (Jun, Seo, et al., 2022; Jun, Birchfield, et al., 2022). Consider a line such as `dplyr::filter(df, x > 0)`, which has a precise, unambiguous meaning as code: keep rows where `x` is positive. As an analytical decision, however, that same line could represent restricting the analysis to a subpopulation for which `x > 0` is substantively meaningful, or excluding observations flagged as invalid by an encoding convention where $x \leq 0$ denotes missingness; nothing in the `filter()` syntax gives any indication of which of these two decisions is being made. Tools such as `tidyverse` (Wickham et al., 2019) have worked to align code more closely with the action being performed, naming operations after what they do rather than how they are implemented. But knowing the action a line of code performs is not the same as knowing the decision it encodes: an analysis requires additional context that the action alone does not carry.

This suggests that a more effective representation of an analysis would be one that describes the decisions made and their rationale, allowing readers to understand the analysis without needing to parse the code itself or reconstruct the rationale from the manuscript. Recent advances in Large Language Models (LLMs) for code generation make such a representation practical: LLMs are now widely used for code generation tasks, demonstrating strong capabilities in translating natural language descriptions into executable programs (Chen et al., 2021; Huang et al., 2022; Jiang et al., 2026; Lai et al., 2023; Shah et al., 2026). Once a decision is described in a schema, an LLM can generate and validate the corresponding code directly from it.

This same representation also makes extending an analysis easier. Data analysis is often underdetermined, with many plausible choices regarding data preprocessing, transformation, aggregation, and model specification (Bertran et al., 2026; Gao & Xiao, 2026; Luo et al., 2025), and a reader may reasonably want to know whether a conclusion would still hold under one of these other choices, for example, "What if we had included observations with $x \leq 0$?" Because each decision is recorded explicitly rather than embedded in code, a reader can specify such an alternative directly and have it evaluated through the same schema and the same LLM-based translation into code, rather than modifying the original script by hand.

In this paper, we present a schema designed to describe data analysis in a structured way that captures the decisions and their justifications, as well as the inputs and outputs of each step. We also demonstrate how this schema, coupled with LLMs, can be used to generate code scripts from natural-language descriptions in manuscripts, and explore alternative decision choices. We apply our schema to the many-analysts football study,

where LLMs create the schema from the decision choices reported in the final reports of each team, and generate and validate the corresponding code scripts against the original published results. In the process, we also uncover an inconsistency in one team's published report between the covariate described in the text and the one implied by their results. We further demonstrate how the schema supports specification and exploration of alternative random effect specifications for Team 5.

The paper is organized as follows: In Section 2, we present our methodology, including the schema representation for describing data analysis and the workflow for generating and validating code from the schema by LLMs. In Section 4, we apply our methodology to the many-analysts football study and demonstrate that we can reproduce the results reported in the paper, identify errors in the original reports, and highlight failure modes of current LLMs. Finally, in Section 5, we discuss the implications of our work and potential future directions.

2. Methodology

As traditional reproducibility practices can be insufficient for understanding complex modern data analyses, we have designed two complementary YAML schemas and the associated workflow to better support comprehension and extensibility in applied data analysis. Our workflow is designed to be 1) decision-centric: analyses are represented as a sequence of decisions with explicit rationale, and 2) reader-oriented exploration: readers can inspect, compare, and extend those decisions.

2.1. *A schema for describing data analysis*

We represent a data analysis, $\mathcal{S} = \{A_1, A_2, \cdots, A_n\}$, as a sequence of n analytical step A_i, applied to a dataset $\mathcal{D}$, where each step, A_i, is defined as a tuple:

$$A_i = \langle f_i, \pi_i, \rho_i, \mathcal{I}_i, \mathcal{O}_i, \omega_i \rangle. \tag{1}$$

The components of A_i can be partitioned into two parts: a required triplet (f_i, π_i, ρ_i) and an optional triplet $(\mathcal{I}_i, \mathcal{O}_i, \omega_i)$. The first component of the triplet, $f_i \in \mathcal{F}$ is the *objective* of a step, identifying the problem the step is trying to solve and thereby fixing the space of alternative decisions that could equally address it (formally introduced in Section 2.2). The second component, $\pi_i \in \Pi$, is the *decision* made to achieve f_i: it is a narration, in natural language, of what a corresponding line of code would do, situated in the context of this specific analysis, rather than the code's syntax itself. The term ρ_i is the *justification* provided for choosing the decision π_i over the other alternatives that could have fulfilled the same objective. Structuring this information into a schema involves defining the three components described above for each step. As an example,

consider the description for calculating the Human Development Index (HDI) (United Nations Development Programme, 2025):

1) Scale all four variables (life expectancy, expected years of schooling, mean years of schooling, and GNI per capita) to the [0, 1] range.
2) Combine the two education variables (expected years of schooling and mean years of schooling) into a single education variable.
3) Combine the three dimensions (health, education, and income) into a single index using a geometric mean.

In Step 3, the objective is to combine the three dimensions. While a simple arithmetic mean could be an alternative decision to fulfill this goal, the geometric mean (the decision) is chosen because it penalizes uneven development among the three dimensions (the justification). The full schema for constructing HDI is presented in Listing 1.

Listing 1 An example schema for calculating Human Development Index (HDI)

```yaml
meta:
  type: schema
nodes:
- tag: step-scaling
  objective: how to handle variables in different scales?
  decision: apply min-max scaling to each variable
  rationale: to convert the variables to the same scale for future
    construction of a single index
- tag: step-education
  objective: how to combine the two education-related variables?
  decision: take the average
  rationale: the arithmetic mean allows perfect substitutability
    between the two variables
- tag: step-combine
  objective: how to combine the three dimensions into a single index?
  decision: use the geometric mean
  rationale: the geometric mean penalizes uneven development
    among the three core dimensions
```

The required triplet (f_i, π_i, ρ_i) is already sufficient on its own for a human reader to follow the analytical logic of a step, without needing access to the underlying dataset. The optional triplet $(\mathcal{I}_i, \mathcal{O}_i, \omega_i)$ adds a further layer on top of this: $\mathcal{I}_i$ and $\mathcal{O}_i$ bind the step to concrete variables in the dataset, resolving ahead of time the variable mapping that an LLM would otherwise have to infer at the point of code generation. Populating them requires a dataset to be available, which is why they remain optional rather than required. While the output of a step can be a pseudo object that does not yet exist in the dataset, the inputs for any given step must be within the existing data scope, comprising variables in the original dataset and all outputs from previous steps. Because the schema is meant to specify an analysis before the corresponding code is generated, an output name can refer to a variable that does not yet exist; this is inherent to using the schema as a specification rather than a post-hoc description of existing code. In

practice, $\mathcal{O}_i$ most often names a column added to the evolving dataset via non-standard evaluation, as in `tidyverse`-style pipelines; a small number of steps, such as model fitting, instead produce a distinct object that is not itself a column of $\mathcal{D}$. Formally, we require:

$$\mathcal{I}_i \subseteq \left(\mathcal{D} \cup \bigcup_{j<i} \mathcal{O}_j \right), \quad \forall i \in \{2, \dots, n\}.$$

When the schema is extracted automatically by Large Language Models (LLMs) from text, the inputs and outputs will be inferred by the LLMs, if a dataset is provided. We also introduce a categorical confidence score, ω_i (low, medium, high), defined as the model's confidence that the recorded (f_i, π_i, ρ_i) correctly reflects the source text; ω_i is low precisely when one or more of these components could not be read off the text directly and had to be inferred by the LLM, and can be used to guide the LLM to ask for clarification to consolidate the schema extraction.

Recording a step's decision π_i does not mean it is the only one that could satisfy its objective f_i: a reader may want to revisit the decision made for some of these steps and consider alternatives that address the same objective, a possibility we formalize next.

2.2. *Describing alternatives*

Another analyst following the HDI schema in Listing 1 might wonder whether the reported index would look different under a different decision at some step. Formally, a step in the schema can have a candidate set of alternatives: step i may admit k_i decisions that all satisfy its objective f_i, $\{A_i^1, \dots, A_i^{k_i}\}$, of which the decision π_i recorded in the schema is just one member, A_i^1. An alternative file specifies the remaining candidates, $A_i^2, \dots, A_i^{k_i}$, for a subset of steps $C \subseteq \{1, \dots, n\}$.

The overall structure of an alternative file is shown in Listing 2. The `meta` field at the top identifies the file as an alternative specification (`type: alternatives`), and the file-level `branch` field, either `single` or `multi`, determines how the candidates at each step combine, described below. The file itself is organized as a list, one entry per step in C (`nodes` in the YAML): each entry names its step in the original schema (via `overrides`) and lists the alternative decisions for that step, together with their justifications.

Listing 2 The skeleton of an alternative YAML file, which consists of a `branch` mode and a list of covered steps, each specifying a step in the original schema and its alternative decisions.

```yaml
meta:
  type: alternatives
  branch: single | multi
nodes:
```

```
  - overrides: <step id>
    alternatives:
      - id: <alternative id>
        decision: <...>
        rationale: <...>
      - id: <alternative id>
        ...
  - overrides: <step id>
    alternatives:
      ...
  ...
```

Under `branch: single`, every step is restricted to a single candidate decision. These alternatives and the unchanged steps in the original schema together form one new analysis. This is useful when an analyst wants to see what a different, internally consistent analysis touching several steps at once would look like. Listing 3 fills in the skeleton to construct such an alternative version of the HDI computation, using a harmonic mean to combine the two education variables in Step 2 and an arithmetic mean to combine the three dimensions in Step 3.

Listing 3 A sample alternative file that changes the decisions for Step 2 and Step 3 in calculating Human Development Index (HDI) to use a harmonic mean and an arithmetic mean, respectively.

```
meta:
  type: alternatives
  branch: single
nodes:
  - overrides: step-education
    alternatives:
      - id: harmonic-mean
        decision: take the harmonic mean of the two education variables
        rationale: penalizes disparity between the two schooling measures
          rather than allowing one to compensate for the other
  - overrides: step-combine
    alternatives:
      - id: arith-mean
        decision: use an arithmetic mean
        rationale: a simple average of the three dimensions
```

Sometimes a reader instead wants to test a range of alternatives for one or more steps independently. Under `branch: multi`, the candidates at each covered step, along with that step's original decision, are crossed to produce every combination. In Listing 4, one alternative is specified for the education step (the harmonic mean) and two for the combination step (an arithmetic mean and the minimum of the three dimensions). Together with the original schema, this produces $(1 + 1) \times (2 + 1) = 6$ analyses. More generally, with k_i candidates at each step $i \in C$, crossing all candidates yields $\prod_{i \in C} k_i$

analyses.

Listing 4 A sample alternative file that explores one alternative decision for Step 2 and two for Step 3 in calculating Human Development Index (HDI). Combining these alternatives with the original schema produces a total of six $(1+1) \times (2+1) = 6$ analyses.

```yaml
meta:
  type: alternatives
  branch: multi
nodes:
  - overrides: step-education
    alternatives:
      - id: harmonic-mean
        decision: take the harmonic mean of the two education variables
        rationale: penalizes disparity between the two schooling measures
          rather than allowing one to compensate for the other
  - overrides: step-combine
    alternatives:
      - id: arith-mean
        decision: use an arithmetic mean
        rationale: a simple average of the three dimensions
      - id: min-agg
        decision: use the minimum of the three dimensions
        rationale: a "weakest-link" aggregation that penalizes any single
          lagging dimension more strongly than an average would
```

2.3. *Workflow*

The overall workflow for schema-based data analysis is illustrated in Figure 1. A schema can be created either manually from the template or automatically by an LLM from a natural-language manuscript. Once created, the schema serves as a structured specification that guides a second LLM in generating the code for the analysis. To improve reliability, our design includes a third LLM agent which is responsible for code validation. While the latest LLMs can generate reasonable code based on prompt, they do not consistently achieve perfect accuracy. The validation process runs the generated script in a sandbox environment and if an error occurs, a third LLM agent will be employed to diagnose the script. The validation agent receives the generated code together with the error message and the corresponding line number produced during the execution, and returns a newer version of the script. The revised script is then re-executed in the sandbox and this cycle continues until one of the three conditions is met: the code runs successfully, a maximum number of attempt is reached, or the user terminates the process. As will be show in Section 4, this validation process resolves the majority of R syntax issues encountered during data wrangling and statistical modelling tasks. On top of supporting a single analytical specification, the workflow also allows users to argument a schema with an alternative YAML file. This expands the original schema into a multiverse of parallel schemas, each representing a different sequence of decision choices. The same code generation and validation workflow is then applied independently to each

schema, producing a corresponding set of validation scripts.

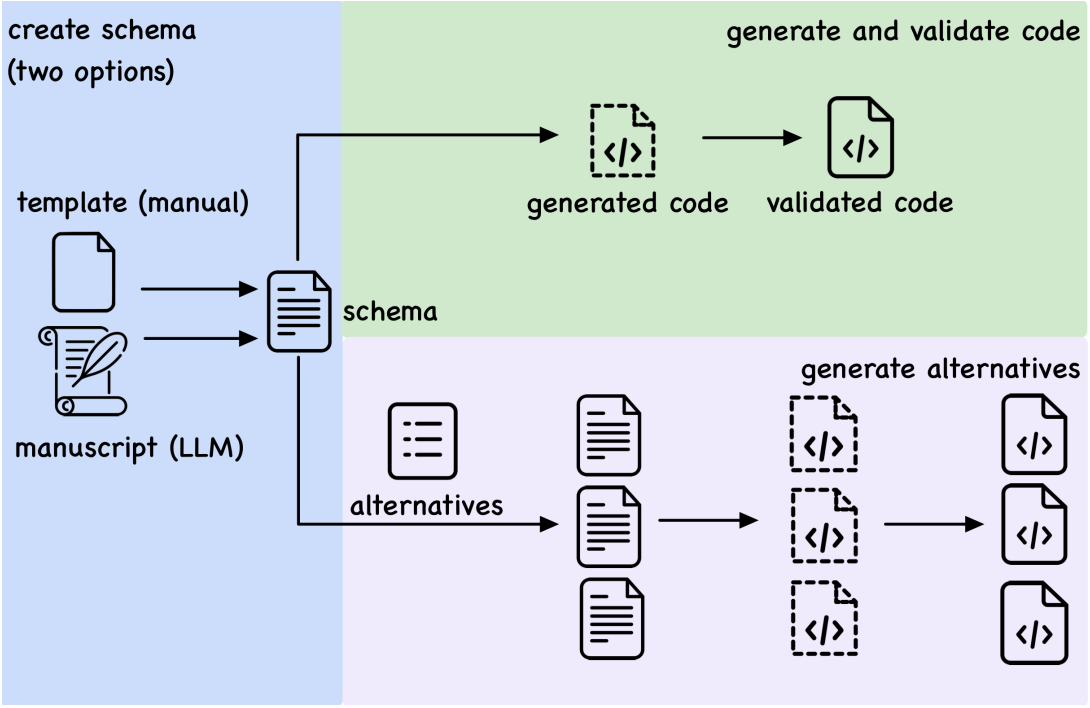


Figure 1. Overview of the schema-driven analysis workflow. Schemas can be authored manually or generated from natural language manuscripts by an LLM. Schemas are used to generate code, which is validated through iterative execution in a sandboxed environment and repaired when errors occur. Alternative specifications can expand a schema into a multiverse of parallel analyses.

Returning to the HDI example, the proposed workflow allows both authors and readers to interact with a data analysis through natural language. The schema in Listing 1 is straightforward to construct from a template and can be included as supplementary material to document the analytical workflow. An author may use the "code generation and validation" path in Figure 1 to automatically generate the analysis script for computing HDI and validate its accuracy. A reader, on the other hand, may wish to explore how the HDI values or country rankings change if the original arithmetic mean (instead of the geometry mean) is used. By combining the original schema in Listing 1 with the alternative specification Listing 4, the "alternative generation" path produces a corresponding analysis script for each alternative with LLMs. In contrast, modifying an existing code script by hand typically requires manually coordinating changes across upstream and downstream code. By representing analyses as a sequence of structured analytical decisions, our workflow allows LLMs to automate much of this process, making it easier to understand the sequence of steps taken in the analysis and to explore alternatives.

We now analyze the uncertainty of the pipeline itself: for a step k extracted by a model M from a source text T, we decompose the total uncertainty of the pipeline at that step into a specification term and an implementation term,

$$V_k^{\text{total}}(M) = V_k^{\text{spec}}(M) + V_k^{\text{impl}}(M),$$

where each component is defined as the expected pairwise divergence, $d(\cdot, \cdot) \in [0, 1]$, between independent runs at the corresponding stage. Specifically, $V_k^{\text{spec}}(M)$ measures variation across repeated schema extractions from T and $V_k^{\text{impl}}(M)$ measures variation across repeated code generations from a single fixed schema. We can further decompose the specification term into an epistemic and an aleatoric component (Hüllermeier & Waegeman, 2021; Kendall & Gal, 2017):

$$V_k^{\text{spec}}(M) = V_k^{\text{epistemic}}(M) + V_k^{\text{aleatoric}},$$

The epistemic component, $V_k^{\text{epistemic}}(M)$, reflects when the model fails to recover a decision that is unambiguously specified in T. The aleatoric component, $V_k^{\text{aleatoric}}$, captures the uncertainty in the source itself: the text T is unclear at step k, independent of M. This distinction has a direct diagnostic use in the quality of the data anlaysis. Persistent dispersion across increasingly capable models suggests that the observed variation reflects ambiguity in the source since a more capable model cannot recover the authors' intent that is not sufficiently expressed in the source text.

3. Application

In this section, we demonstrate the schema workflow from an author's perspective: what it looks like for an author to provide a schema alongside a manuscript, and what that then lets an audience do to question and explore the analysis with much less effort than the manuscript and its code alone would require. We use Yee et al. (2025)'s GAITD regression analysis of heaped smoking data, published in the Annals of Applied Statistics, as the example. Yee et al. (2025) report four separate smoking-related responses; for demonstration purposes we build the schema for only the first, smoking initiation age (P1, Section 5.1 of their paper), and the same workflow extends to the other three.

Listing 5 contains fourteen entries; rather than list them all, we highlight one as representative of what each entry records. Every entry follows the *objective*, *decision*, and *rationale* structure introduced in Section 2.1. Consider `special_value_mid_risk`: the decision is to treat ages 12-14 and 19-21 as one inflated group with extra probability mass, since these ages show a moderate, secondary elevation in the spikeplots, surrounding the sharper peak already attributed to the high-risk group (ages 15-18). Recording the rationale along with the decision allows a reader to agree or disagree with the authors' choice, and to question and explore the analysis, as we do later in this section. The remaining thirteen entries follow the same structure.

Listing 5 A schema for the GAITD regression analysis of smoking initiation age in Yee et al. (2025), constructed from the manuscript and intended to be provided alongside it as a self-contained account of the analysis.

```
meta:
  type: schema
nodes:
- id: complete_case_filtering
  objective: handle missing data in selected variables
  decision: apply complete-case filtering to all columns except TC30, TCquit,
   ↪  and SCA which have NAs by design
  rationale: TC30, TCquit, and SCA are exempted from the completeness check
   ↪  since they are interlaced with NAs by design (e.g., SCA being NA means
   ↪  the person has not yet quit, a valid state rather than missingness);
   ↪  every other selected column is required to be non-missing because most
   ↪  smoking studies adjust for age, marital status, education and race, and
   ↪  this analysis follows suit.

- id: age_restriction
  objective: define the age range for analysis
  decision: restrict analysis to respondents aged less than 80 years (age <
   ↪  80)
  rationale: age was right-censored at 80

- id: data_quality_correction
  objective: handle logically impossible self-reported values
  decision: set both SCA and SIA to NA where SCA - SIA < 0
  rationale: quitting age before smoking initiation age is a logical
   ↪  impossibility and such values cannot be trusted

- id: impute_marital_young
  objective: assign a defensible marital status for young respondents,
   ↪  regardless of what was recorded
  decision: recode marital status to "Never married" for everyone age < 20
  rationale: this age group is effectively certain to have never been
   ↪  married, so the value is set by age rather than trusting the raw survey
   ↪  response

- id: impute_education_young
  objective: assign a defensible education category for young respondents,
   ↪  regardless of what was recorded
  decision: recode education to "< 9th grade" for age < 14; "9-11th grade"
   ↪  for 14 <= age <= 16; "High school" for 17 <= age <= 19
  rationale: for respondents still of school age, highest-grade-completed is
   ↪  essentially determined by age, so the value is set by age rather than
   ↪  trusting the raw survey response

- id: define_ever_smokers_sample
  objective: define the analytic sample of ever-smokers
  decision: include as valid observation if both conditions are met -- 1) SIA
   ↪  is not missing, and 2) either SCA and TCquit are both not missing, or
   ↪  SCA is missing and TC30 is not missing
```

```
  rationale: SCA is missing by design for anyone who has not yet quit. the
   ↪ two-branch rule instead admits ex-smokers through the SCA/TCquit branch
   ↪ and current smokers through the TC30 branch

- id: parent_distribution_choice
  objective: select appropriate probability distribution for count response
   ↪ variable
  decision: adopt negative binomial distribution as the parent distribution
  rationale: the SIA distributions iboth unimodal and right-skewed, so a NBD
   ↪ parent is reasonable; more generally, there is evidence of
   ↪ overdispersion relative to a Poisson distribution.

- id: special_value_mid_risk
  objective: specify how to model inflation at ages surrounding peak teenage
   ↪ smoking uptake
  decision: treat ages 12, 13, 14, 19, 20, 21 as a parametrically-inflated
   ↪ group, with the extra probability mass there sharing the same
   ↪ distributional shape (mean and dispersion) as the main population
  rationale: this mid-risk subgroup represents early and later smokers
   ↪ surrounding the high-risk age subgroup; these ages showed a moderate,
   ↪ secondary elevation in repeated training/test spikeplots of the data

- id: special_value_high_risk
  objective: specify how to model inflation at peak teenage smoking uptake
   ↪ ages
  decision: treat ages 15, 16, 17, 18 as a nonparametrically-inflated group,
   ↪ with each age given its own freely-estimated extra probability
  rationale: this high-risk subgroup corresponds to frenetic teenage smoking
   ↪ uptake, the age range where the largest number of regular smokers
   ↪ start; these ages showed the largest, most persistent spikes across
   ↪ repeated training/test spikeplots of the data.

- id: alteration_operator_decision
  objective: determine whether to apply alteration operators
  decision: do not use alteration operators (A.mix and A.mlm set to NULL)
  rationale: not needed for this model specification

- id: deflation_operator_decision
  objective: determine whether to apply deflation operators
  decision: do not use deflation operators (D.mix and D.mlm set to NULL)
  rationale: not needed for this model specification

- id: truncation_decision
  objective: determine whether to specify a truncated region
  decision: do not use truncation (no truncated region specified for SIA)
  rationale: truncation not appropriate for this response variable

- id: covariate_selection
  objective: select covariates for the linear predictor of parent
   ↪ distribution mean
  decision: include race2, gender, marital, and educ as covariates
```

```
  rationale: most smoking studies adjust for age, marital status, education
    ↪ and race

- id: fit_final_model
  objective: fit the final GAITD regression model
  decision: fit a single vglm that puts together the pieces above --
    ↪ response, parent distribution, special-value structure (inflation,
    ↪ alteration, deflation, and truncation), and covariates. The covariates
    ↪ only shift the mean of the negative binomial; the dispersion and the
    ↪ mid-/high-risk inflation probabilities are held fixed across the sample
  rationale: only the mean is modeled with covariates; the dispersion and
    ↪ inflation probabilities are kept intercept-only, since extending
    ↪ covariate effects to them would substantially increase the number of
    ↪ parameters to estimate and make the results harder to interpret
```

As a check that this schema is accurate, generating and validating code from it produces the R script in Listing 6, which reproduces the coefficients Yee et al. (2025) report for smoking initiation age exactly, matching their Table 2. This check is useful beyond validating our own workflow: a schema that reproduces a published result could accompany the manuscript as reproducibility material in its own right, alongside code and data availability, and a reader can trust the schema in Listing 5 to accurately describe the analysis it documents.

Listing 6 The R script generated and validated from the schema in Listing 5, annotated with the `id` of the schema step each block implements.

```r
library(VGAMdata)
library(VGAM)
library(dplyr)

data("smqP", package = "VGAMdata")

cols <- c('TC30', 'TCquit', 'SCA', 'educ', 'marital',
          'SIA', 'SEQN', 'age', 'gender', 'race2',
          "passiveSmoke_job", "passiveSmoke_rest",
          "passiveSmoke_bar", "passiveSmoke_car",
          "passiveSmoke_home", "passiveSmoke_other",
          "passiveSmoke_ecigarettes")
df <- smqP |> select(all_of(cols))

# Step 1: complete_case_filtering
# Apply complete-case filtering to all columns except TC30, TCquit, and SCA
cols_require_complete <- setdiff(cols, c("TC30", "TCquit", "SCA"))
df_complete <- df |>
  filter(complete.cases(pick(all_of(cols_require_complete))))

# Step 2: age_restriction
# Restrict analysis to respondents aged less than 80 years
df_age_restricted <- df_complete |> filter(age < 80)
```

```r
# Step 3: data_quality_correction
# Set both SCA and SIA to NA where SCA - SIA < 0
df_age_restricted <- df_age_restricted |>
  mutate(
    SCA_corrected = if_else(!is.na(SCA) & !is.na(SIA) & (SCA - SIA < 0),
      ↪ NA_integer_, SCA),
    SIA_corrected = if_else(!is.na(SCA) & !is.na(SIA) & (SCA - SIA < 0),
      ↪ NA_integer_, SIA)
  )

# Step 4: impute_marital_young
# Recode marital status to "Never married" for everyone age < 20
df_age_restricted <- df_age_restricted |>
  mutate(
    marital_imputed = if_else(age < 20, "Never married",
      ↪ as.character(marital))
  )

# Step 5: impute_education_young
# Recode education based on age for young respondents
df_age_restricted <- df_age_restricted |>
  mutate(
    educ_imputed = case_when(
      age < 14 ~ "< 9th grade",
      age >= 14 & age <= 16 ~ "9-11th grade",
      age >= 17 & age <= 19 ~ "High school",
      TRUE ~ as.character(educ)
    )
  )

# Step 6: define_ever_smokers_sample
# Include if: SIA not missing AND (both SCA and TCquit not missing OR SCA
  ↪ missing and TC30 not missing)
df_ever_smokers <- df_age_restricted |>
  filter(
    !is.na(SIA_corrected) &
    ((!is.na(SCA_corrected) & !is.na(TCquit)) | (is.na(SCA_corrected) &
  ↪ !is.na(TC30)))
  )

df_ever_smokers <- droplevels(df_ever_smokers)

# Step 7: parent_distribution_choice
# Adopt negative binomial distribution as the parent distribution
parent_distribution <- negbinomial

# Step 8: special_value_mid_risk
# Ages 12, 13, 14, 19, 20, 21 as parametrically-inflated group (i.mix)
i_mix_spec <- c(12, 13, 14, 19, 20, 21)
```

```r
# Step 9: special_value_high_risk
# Ages 15, 16, 17, 18 as nonparametrically-inflated group (i.mlm)
i_mlm_spec <- c(15, 16, 17, 18)

# Step 10: alteration_operator_decision
# Do not use alteration operators
a_mix_spec <- NULL
a_mlm_spec <- NULL

# Step 11: deflation_operator_decision
# Do not use deflation operators
d_mix_spec <- NULL
d_mlm_spec <- NULL

# Step 12: truncation_decision
# Do not use truncation
truncation_spec <- NULL

# Step 13: covariate_selection
# Include race2, gender, marital, and educ as covariates
covariate_formula <- SIA_corrected ~ race2 + gender + marital_imputed +
   ↪  educ_imputed

# Step 14: fit_final_model
# Fit a single vglm with response, parent distribution, special-value
   ↪  structure, and covariates
fitted_model <- vglm(
  covariate_formula,
  family = gaitdnbinomial(
    i.mix = i_mix_spec, i.mlm = i_mlm_spec,
    a.mix = a_mix_spec, a.mlm = a_mlm_spec,
    d.mix = d_mix_spec, d.mlm = d_mlm_spec,
    truncate = truncation_spec
  ),
  data = df_ever_smokers
)
```

Beyond confirming the schema's accuracy, this workflow lets a reader write an alternative file directly against the schema and obtain generated, validated code, rather than going through the authors' original script and modifying it by hand. Yee et al. (2025)'s supplementary material reports exactly this kind of question from a reviewer, who suggested splitting the mid-risk group into two, keeping ages 12-14 as inflated but treating ages 19-21 as altered instead, since that older subgroup shows a qualitatively different, declining pattern of uptake. Listing 7 encodes the reviewer's suggestion as an alternative file, overriding three steps at once: `special_value_mid_risk` narrows the inflated group to ages 12-14, `alteration_operator_decision` turns on an alteration group for ages 19-21, and `covariate_selection` drops to an intercept-only model, matching how the authors themselves computed AIC for this comparison; `branch: single` ties the

three overrides into one refit, rather than crossing them into separate branches.

Listing 7 An alternative YAML file specifying a joint change to the mid-risk inflation, alteration, and covariate steps in the GAITD schema, treating ages 19-21 as an alteration group rather than pooling them into the mid-risk inflated group.

```yaml
meta:
  type: alternatives
  branch: single
nodes:
  - overrides: special_value_mid_risk
    alternatives:
      - id: ap1921-special-value
        decision: "treat ages 12, 13, 14 only as a parametrically-inflated
   ↪  group, with the extra probability mass there sharing the same
   ↪  distributional shape (mean and dispersion) as the main population; ages
   ↪  19, 20, 21 are no longer pooled into this group (see the alteration
   ↪  operator decision instead)"
        rationale: "ages 19-21 mark less frenetic, declining smoking uptake
   ↪  as respondents age out of the high-risk window, a qualitatively
   ↪  different pattern from the early uptake at ages 12-14, so they are
   ↪  moved out of this pooled inflated group"
  - overrides: alteration_operator_decision
    alternatives:
      - id: ap1921-alteration
        decision: "use ages 19, 20, 21 as an alteration group (A.mix) instead
   ↪  of leaving alteration operators unused"
        rationale: "ages 19-21 mark less frenetic, declining smoking uptake
   ↪  as respondents age out of the high-risk window, a qualitatively
   ↪  different pattern from the early uptake at ages 12-14, so they are
   ↪  represented as an alteration rather than folded into the same inflated
   ↪  group"
  - overrides: covariate_selection
    alternatives:
      - id: ap1921-covariates
        decision: "do not include any covariates (intercept-only model)"
        rationale: "this comparison is about which special-value structure
   ↪  fits better, so covariates are left out to compare the two structures
   ↪  on equal footing, the same way the special-value structure was first
   ↪  established on an intercept-only model before covariates were added"
```

Generating and validating code from this alternative, without the reader needing to touch the original script by hand, reproduces the AIC and KLD values the reviewer's refit is reported to have achieved in the supplementary material (AIC = 19340.36, KLD = 0.380). What took the original authors a manual refit and a paragraph in the supplement is, for a reader equipped with this schema and its alternative file, a single generation-and-validation step.

4. Application: many-analysts football study

In this section, we apply our methodology to the many-analysts football study by Silberzahn et al. (2018) to reproduce the reported estimated effect size by each team. During this process, we identify some errors in the original reports and highlight failure modes of current LLMs.

Silberzahn et al. (2018) invited 29 research groups to conduct the same data analysis, answering whether referees are more likely to give red cards to dark-skin-toned players than to light-skin-toned players. The provided dataset contains two raters' skin color ratings of players' pictures, the aggregated number of red cards received by a player from a referee across all matches in that player-referee pair, and other relevant player, referee, and game characteristics: player birthday, height, weight, player position, league country, number of victories, ties, and defeats, and number of yellow and red cards received. In Silberzahn et al. (2018), each group first conducted an initial analysis separately, followed by a round-robin peer evaluation. Each group was then given the opportunity to incorporate feedback from the discussion and submitted a final analysis as a report with a formal Method section. All the materials related to the study are publicly available on the Open Science Framework (OSF).

Based on the reports presented by each team, most participating groups adopted a modelling approach to estimate the effect of skin color rating on the likelihood of receiving a red card. As an illustrative example, Group 5 first transformed the data into player–game observations, resulting in a binary response variable for whether a player received a red card, and removed observations with missing values using list-wise deletion. The resulting data were modelled using a generalized linear mixed model (GLMM) with a binomial error distribution. The average skin color rating from two raters was used as the primary covariate, and its estimated coefficient was the main parameter of interest. To account for unobserved heterogeneity, the model included random effects for player and referee, as well as the average skin color rating across referee's country of origin. While Group 5's analysis illustrates one possible approach, other groups may adopt different data preprocessing decisions, distributional assumptions (e.g., linear, logistic, or Poisson models), or specifications of fixed and random effects, or may conduct the analysis in a Bayesian framework.

Team reports are not straightforward to use as input directly. Beyond the analysis of interest, whether skin color rating predicts red cards, the many-analysts study also includes a second research question and report confidence interval calculations for the estimated effect, neither of which is relevant to the schema extraction or code generation we study here. The report structure adds a further complication: some teams write the final analysis as edits against the initial analysis, for example a covariate dropped, a variable rescaled, a different outcome, rather than as a self-contained methodology section. Recovering a single, self-contained methodology description while setting aside the unrelated research question is its own nontrivial comprehension problem, separate from the task of generating the schema and analysis code that we study here. We

therefore pre-extract a simplified methodology text for each team and use this, rather than the original report, as the input for our simulation.

We send the text description of each team to an LLM to generate the schema, and then generate the code script from the schema. The code script is then run by a separate LLM to validate it for potential syntax errors.

To accelerate the evaluation of each script, we manually replace, in place, the `lm()`/`glm()` command with `fixest::feols()`/`fixest::feglm()` (Bergé et al., 2026) for generalized linear models, and the `lme4::glmer()` (Bates et al., 2015) function with `glmmTMB::glmmTMB()` (Brooks et al., 2017) for generalized mixed-effects models, after the scripts are generated but before validation. These two functions are in-place replacements of the standard functions for estimating the relevant models, with multi-threading in C++ (Bergé et al., 2026) and Template Model Builder (TMB) (McGillycuddy et al., 2025) for better performance. This replacement should not affect the estimation of the effect size. All the LLM-related simulations are conducted with Claude-Opus-4.5.

Of the 29 teams, we report results for 25. The remaining four (Teams 7, 12, 17, and 18) specify analyses that are too computationally expensive to run inside an iterative validate-and-repair loop, regardless of whether the schema or generated code is correct — MCMC-based hierarchical models with thousands of random effects, a nonparametric Bayesian clustering procedure, and a fixed-effects model with several thousand parameters. Across the 25 teams we reproduce, all team scripts were successfully executed after the validation phase. Notably, two scripts ran successfully without any validation, whereas the vast majority reached completion after two or three iterations. The most frequent correction involved resolving broken and overflowing syntax in comments.

Figure 2 shows the comparison of the estimated effect sizes, in odds ratio, from our workflow (simulated) and those reported in Silberzahn et al. (2018) (published). The odds ratio is well recovered for most teams, though six show a noticeably larger gap: Team 1, 9, 15, 21, 23, and 27. Looking closely at each team's report and generated script, the causes split into two stages of our pipeline, plus one case that has nothing to do with the LLM at all.

For Team 27, the report is genuinely ambiguous: the authors write: "In addition to the independent variables specified in the initial approach, … a variable for player position was included, as well as a variable for the league of the player." A statistician reading this might reasonably take it as shorthand for extending the model with interactions, consistent with how the other covariates entered the initial model, but the LLM read it literally and added only main effects. Team 9 fails for a similar reason at the same stage: the text names "referee country" as the covariate, yet the reported estimation table lists "player country" instead, and the LLM followed the text rather than reconciling it against the table. In both cases the schema itself is already wrong before any code is written, so we treat these as schema-extraction errors.

Team 1 and Team 23, by contrast, get the schema right but the code wrong. Both require reshaping the data from player-referee pairs to individual games, disaggregating

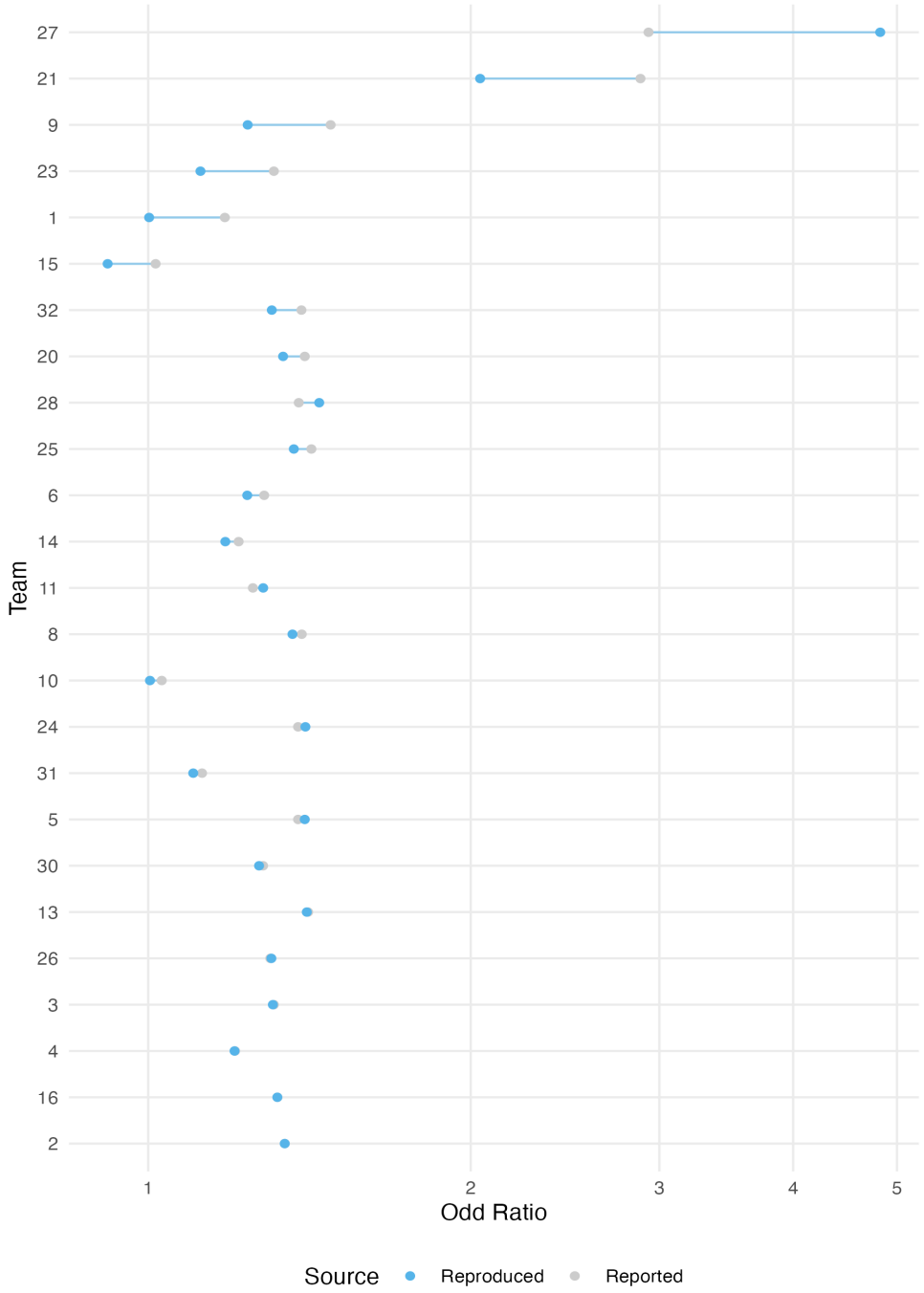


Figure 2. Comparison of the odds ratio of the effect of skin color on receiving red cards in football games from our schema-based reproduction (reproduced) vs. the effect size reported in Silberzahn et al. (2018) (reported). Teams are ordered by the absolute difference between the reproduced and reported odds ratio. The reproduction recovers the reported odds ratio closely for most teams; the largest deviations occur for Team 1, 9, 15, 21, 23, and 27.

win/loss/draw counts into game-level records. The schema correctly specifies this step, but the generated code fails to execute it as intended, and the resulting error passes through validation silently rather than triggering a repair.

Team 15's discrepancy is unrelated to either stage: it comes down to a difference in default contrast coding between R (our reproduction) and SPSS (the original analysis) — SPSS's effect coding compares each level to the grand mean, while R's default dummy coding compares to a reference level. Setting the `contrast` option in R to match recovers the reported odds ratio exactly, so this reflects a software convention rather than a modeling or LLM error.

That leaves Team 21, whose discrepancy we have not yet traced to a specific cause [TODO: resolve before submission]. With that exception, the pattern across the remaining five teams is that LLM error is split evenly between the schema-extraction and code-generation stages, while the sixth discrepancy is a reproduction artifact rather than a model failure.

4.1. *Exploring alternatives*

During the modelling process, researchers may consider a range of alternative specifications, such as different covariate and random effects structures, before arriving at a final model. This can be seen as a set of alternative decisions within the same modelling step, which fits naturally into the schema workflow described in Section 2.2. As an example, we use Team 5 from the football study above to demonstrate this. Team 5 uses a binomial generalized linear mixed model (GLMM) to estimate the effect of skin tone rating on the number of red cards received using a logit link function, but tests different random effect structures:

$$\textbf{Model 1 (gm1):} \quad \begin{aligned} \text{logit}(p_{ijk}) &= \beta_0 + \beta_1 X_{ijk} + u_j + v_k \\ u_j &\sim N(0, \sigma^2_{\text{player}}), \quad v_k \sim N(0, \sigma^2_{\text{ref}}) \end{aligned}$$

$$\textbf{Model 2 (gm2):} \quad \begin{aligned} \text{logit}(p_{ijk}) &= \beta_0 + \beta_1 X_{ijk} + u_j + v_{0k} + v_{1k} X_{ijk} \\ u_j &\sim N(0, \sigma^2_{\text{player}}), \quad \mathbf{v}_k \sim N\left(\mathbf{0}, \Sigma_{\text{ref}}\right) \end{aligned}$$

$$\textbf{Model 3 (gm3):} \quad \begin{aligned} \text{logit}(p_{ijkm}) &= \beta_0 + \beta_1 X_{ijkm} + u_j + v_k + w_{0m} + w_{1m} X_{ijkm} \\ u_j &\sim N(0, \sigma^2_{\text{player}}), \quad v_k \sim N(0, \sigma^2_{\text{ref}}), \quad \mathbf{w}_m \sim N\left(\mathbf{0}, \Sigma_{\text{country}}\right) \end{aligned}$$

where p denotes the probability of a player receiving a red card, and X represents the player's skin tone rating. The subscripts i, j, k, and m index the individual observation, player, referee, and the referee's home country, respectively. The parameter β_0 represents the global baseline log-odds of receiving a red card, and β_1 estimates the fixed effect of

a player's skin tone. The random effects include

- a random intercept for players, $u_j \sim N(0, \sigma^2_{\text{player}})$, in Models 1, 2, and 3,
- a random intercept for referees, $v_k \sim N(0, \sigma^2_{\text{ref}})$, in Models 1 and 3, plus an additional random slope for referees in Model 2, and
- a random intercept and slope for referee countries, $\mathbf{w}_m \sim N\left(\mathbf{0}, \Sigma_{\text{country}}\right)$, in Model 3.

These alternatives are specified in a separate YAML file, following the format described in Section 2.2 and shown in Listing 8. This file is then merged with the original schema to produce a separate schema for each alternative for code generation and validation. The validated scripts are then executed to obtain the estimated random effects in Table 1. This result matches the random effects reported in the team report, subject to rounding errors. Similarly, the model fitting statistics can be extracted from the estimation for selecting the best fit.

Listing 8 Example of a YAML file specifying alternatives for the random effects structure in the many-analysts analysis of Group 5.

```yaml
meta:
  type: alternatives
  branch: multi
nodes:
- overrides: specify_random_effects_structure
  alternatives:
  - id: gm1
    decision: a random effect of player and referee
    rationale: ''
  - id: gm2
    decision: a random effect of player and referee, and a random effect of
    ↪  skin-tone across referees
    rationale: ''
  - id: gm3
    decision: a random effect of player, referee, and referees' countries of
    ↪  origin, and a random effect of skin tone across referees' countries of
    ↪  origin
    rationale: ''
```

Table 1. Summary of random effect estimates for Group 5. The three models (gm1, gm2, and gm3) represent different specifications of the random effects structure to account for the non-nested cross-classified data structure. Cells are NA where a given random effect is not part of that model's specification.

Group	Term	gm1	gm2	gm3
Player	Intercept	0.3737	0.3738	0.2899
Referee	Intercept	0.1528	0.1932	0.0748
Referee	Referee rating	NA	0.0286	NA
Referee's country	Intercept	NA	NA	0.3990

Table 1. Summary of random effect estimates for Group 5. The three models (gm1, gm2, and gm3) represent different specifications of the random effects structure to account for the non-nested cross-classified data structure. Cells are NA where a given random effect is not part of that model's specification.

Group	Term	gm1	gm2	gm3
Referee's country	Referee rating	NA	NA	0.0870

5. Discussion

Our results highlight a persistent gap between computational reproducibility and analytical understandability in modern data analysis workflows. While current reproducibility practices ensure that published results can be regenerated from shared data and code, as analyses grow in complexity and involve iterative decision-making processes, they are less of an ideal for readers to understand. Our proposed schema addresses this gap by introducing an intermediate representation that explicitly encodes analytical decisions to bridge between narrative descriptions and implementation details. When coupled with LLMs, this representation enables the extraction of analytical structure from written reports and its translation into analysis scripts, and supporting the exploration of alternative analytical choices.

Our approach differs in this respect from recent work that builds fully automated systems using multiple LLM agents to plan, generate, execute, and refine data analyses (Hu et al., 2024; Jansen et al., 2025). These systems rely on the same underlying capability of translating analytical specifications into code that our workflow uses, but they aim to automate the analytical decisions themselves, not just their translation into code. The failure modes we document in Section 4 suggest this is premature: even the comparatively narrower task of translating an already-specified decision into working code is not yet fully reliable. We take this as support for our position that the harder and more consequential part of a data analysis is the sequence of decisions a researcher makes, not the code that implements them, and that this part should remain explicit and in the hands of the analyst rather than be automated away.

To further examine how this workflow behaves in practice, we apply it to the many-analysts football study. This analysis allows us to identify errors in the original reports and to characterize failure modes of current LLMs to generate code for data science. Beyond validating the workflow itself, the schema-driven reanalysis catches a real inconsistency in a published report (Team 9's text-table mismatch, discussed in Section 4), which without an explicit account of the underlying decisions, would not have caught through computational reproducibility alone. Decomposing the largest discrepancies between our reproduced and reported effect sizes reveals that LLM error arises from two distinct stages of the pipeline, in roughly equal measure: mis-extraction of the schema from an ambiguous report, and correct schema extraction followed by incorrect code generation. This distinction matters for where future effort should be directed

— schema-extraction errors call for better disambiguation strategies (e.g., using the confidence score ω_i to prompt for clarification, as introduced in Section 2), whereas code-generation errors call for improvements to the validation stage itself.

For the schema-extraction errors specifically, we find that LLMs require a fairly precise specification in order to generate the code correctly for the intended random effect structure. For an educated statistician, the phrase "a random effect of referee number across average rating" would typically imply both a random intercept and a random slope. However, we observe that LLMs interpret this specification literally as a random slope only. This suggests the gap between the latest models and human analysts in specialized statistical knowledge (Sun et al., 2026).

Our validate-and-repair loop assumes each candidate script can be executed and diagnosed within an interactive timeframe. This assumption does not hold for analyses that are too computationally expensive to run inside such a loop (Teams 7, 12, 17, and 18 in the many-analysts football study). For future direction, this could be addressed by implementing a background job queue to allow for longer-running scripts to be executed asynchronously.

For the code-generation errors, the LLM successfully corrects a range of implementation issues during validation, including errors caused by overflowing comments and other syntax issues. However, we also observe challenges in data reshaping tasks when the transformation requires changes in the unit of analysis. In the study, some teams decide to transform the data from the original aggregated player-referee dyad-level format to game-level format. This transformation requires disaggregating aggregated outcome counts (win, loss, and draw) into individual game records, changing the unit of analysis from dyads to matches. We also observe some difficulties in parsing birthdates. Since the LLM does not have access to the underlying raw data, due to privacy constraint, it must rely on trial and error when inferring the correct date formats ("DD-MM-YYYY."). This suggests that certain types of data analysis tasks may not be fully automatable in settings where the model cannot directly inspect the data, even if the data is theoretically available for computation.

While the schema requires a justification ρ_i for each decision, nothing in the format itself can guarantee that this justification is substantive rather than perfunctory; an analyst can satisfy the schema with a low-effort entry in the same way that a required code comment or commit message can be filled with boilerplate. The schema makes recording a decision's rationale cheap and structured, but, as with any documentation requirement, cannot enforce that authors use it thoughtfully. It remains an open question whether resolving $\mathcal{I}_i$ and $\mathcal{O}_i$ to concrete dataset variables ahead of time, rather than leaving this mapping for the LLM to infer at the point of code generation, meaningfully improves the reliability of the resulting code. [maybe we want a controlled experiment to compare code generation with or without pre-resolved input and output variables]

The reproducibility-understandability issue considered in this work is still at an early stage, some software decision choices are worth examining. We currently implement a

prototype as an R package, although the understandability problem is broadly applicable to statistical analyses and is not associated to any particular programming language. Hence, we envision the implementation eventually evolving into a general purpose tool, such as a command-line utility, that can support analyses across different programming languages. In the current implementation, analytical decisions are recorded in a single YAML file. An alternative would be to record individual decisions in separate Markdown files. While YAML files provide an explicit structure for machine readability, markdown files offer greater flexibility for documenting decisions. Recording decisions in separate files would also allow files to be linked across time to understand how decisions evolve or change over the course of an analysis and can potentially support a broader research-management system if the decisions are properly attributed to individual researchers and maintains a record of the changes leading to the final analysis.

6. Acknowledgments

The implementation of the schema is available as an R package, `tines`, available on CRAN. The source code for reproducing the work reported in this paper is available in the Supplementary Materials and can be found at: https://github.com/huizezhang-sherry/paper-schema. The article is created using Quarto (Allaire et al., 2026) in R (R Core Team, 2026).

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